Navaneeth S

AI/ML Engineer | Data Scientist | Generative AI & RAG
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LinkedIn | GitHub | Portfolio

SUMMARY

AI/ML Engineer with an M.Tech in AI & ML. Built and deployed a production RAG assistant at GE Aerospace (LangChain, LangGraph, Amazon Bedrock, pgvector on AWS) serving 10-12K monthly users and resolving ~92% of queries without a ticket. First-author researcher in deep learning for SAR imagery (under review, Remote Sensing Letters), with hands-on experience in hybrid-search RAG and agentic AI systems. Seeking full-time GenAI/LLM, ML Engineer, or Data Science roles. Available immediately.

PROFESSIONAL EXPERIENCE

Data Science Intern — GE Aerospace **Jan 2026 – Jul 2026**
Bangalore, India (Hybrid)

- Designed and deployed a **production-grade RAG assistant** (LangChain, LangGraph, Amazon Bedrock, PostgreSQL/pgvector, AWS Lambda, S3) for travel, Source-to-Pay and indirect-procurement queries, available to ~50K employees with **10–12K monthly users**. Owned ingestion, chunking, retrieval, prompting, evaluation and deployment.
- Indexed **3,200+ policy documents** (PDF, Word, Excel, HTML, PowerPoint); answers in 2–12 seconds, **resolving ~92% of queries without a ticket**. Added Amazon Bedrock Guardrails and tracked answer quality through user feedback.
- Automated escalation for unresolved travel queries into ticket creation; **travel ticket resolution SLA improved from 15 to 5 days**, saving an estimated **5,000 labour hours per month**.
- Built data extraction pipelines and workflow automation (Python, pandas, LLMs) for an **AI-driven FP&A forecasting platform** for Corporate Functions Purchased Services, part of an initiative targeting a **50% reduction in ~5,500 annual manual hours**.

RESEARCH PUBLICATION

Structural–Radiometric Feature Fusion Using Multi-Scale SVD Profiles for Urban Flood Segmentation in SAR Images **Under Review — 2026**
Navaneeth S, Dr. Akshara P Byju | Remote Sensing Letters

- Designed a novel **MSVDP (Multi-Scale SVD Profile) dual-encoder cross-attention U-Net** fusing 8-channel multitemporal SAR (VV/VH intensity and coherence) with a 9-channel SLIC-superpixel SVD structural tensor (3 descriptors × 3 scales) for 3-class urban flood segmentation.
- Achieved **0.649 ± 0.006 mIoU and 98.77% pixel accuracy** on UrbanSARFloods, outperforming nine baselines (incl. U-Net++, ResUNet, MANet) — +12.9 points over U-Net and a statistically significant gain over the strongest baseline ($p = 0.041$).
- Built the complete pipeline in PyTorch: SLIC multi-scale feature extraction, dual-stream residual encoders, cross-attention fusion and a composite Focal + Dice + Tversky + Cross-Entropy loss to handle severe class imbalance.

KEY PROJECTS

DocuMind RAG — Enterprise Document Q&A System GitHub **RAG · pgvector · FastAPI**

- Built a production-style **RAG system** for grounded Q&A over enterprise documents: hybrid retrieval (pgvector dense + PostgreSQL full-text) fused with Reciprocal Rank Fusion, cross-encoder reranking and cited, streamed answers via FastAPI and Streamlit.
- Added prompt-injection guardrails, confidence-based fallbacks and cost/latency telemetry; evaluated on a golden set (**Precision@5 0.95, faithfulness 0.98**); Dockerized and deployed on Render.

AgentIQ — Agentic Data-Analysis Platform GitHub **LangGraph · Agentic AI · Text-to-SQL**

- Built a **LangGraph planner–executor–reflector agent** that turns natural-language questions into SQL-backed findings, charts and downloadable HTML reports over SQLite or uploaded CSVs, with bounded replanning on failures.
- Exposed full execution evidence (plans, tool calls, tokens, cost) with multi-provider LLM support (Groq, Anthropic, OpenAI-compatible); FastAPI + Streamlit, Docker, **45 automated tests** in CI, live demo on Render.

SignAI — Real-Time Sign Language Interpreter GitHub **Computer Vision · MediaPipe · Flask**

- Engineered a real-time sign language recognition system delivering gesture-to-text and speech conversion at **under 100 ms latency** using OpenCV and MediaPipe.
- **Finalist at Tredence Infinity AI Hackathon 2025** — demonstrated a live prototype to 500+ participants.

EDUCATION

M.Tech in Computer Science & Engineering — Artificial Intelligence & Machine Learning — **2024 – 2026**
Amrita School of Computing, Amritapuri — Amrita Vishwa Vidyapeetham — CGPA: 8.91 / 10

B.Tech in Computer Science and Engineering — **2020 – 2024**
College of Engineering Kallooppara — APJ Abdul Kalam Technological University — CGPA: 7.68 / 10

Senior Secondary — **2020**
Placid Vidya Vihar, Changanssery — CBSE — 80.2%

Secondary — **2018**
SBHSS, Mannar — CBSE — 92.2%

TECHNICAL SKILLS

Generative AI & LLMs	LangChain, LangGraph, RAG, Prompt Engineering, Fine-Tuning, AI Agents, Agentic AI, Transformers, Amazon Bedrock, Hybrid Search
ML & Deep Learning	PyTorch, TensorFlow, Scikit-learn, LSTM, Computer Vision, Semantic Segmentation, Cross-Attention
Cloud & MLOps	AWS (Lambda, S3, Bedrock), Docker, Git, GitHub, CI/CD, Serverless Architecture, Streamlit
Data & Databases	PostgreSQL, pgvector, MongoDB, Vector Databases, Pandas, NumPy, SQL
Programming	Python, Java, SQL
Frameworks & Tools	FastAPI, Flask, OpenCV, MediaPipe

CERTIFICATIONS & ACHIEVEMENTS

- **IBM Generative AI Engineering Professional Certificate** — IBM / Coursera: 16-course program covering LLMs, RAG, LangChain, fine-tuning and AI agents.
- **Google Build with AI Bootcamp** — Google Developers: hands-on AI agent development, workflow architecture and GenAI production integration.
- **Finalist** — Tredence Infinity AI Hackathon 2025 and Kochi Hackathon 2025.
- **NPTEL — Introduction to Large Language Models** — NPTEL online course on LLM fundamentals.

LANGUAGES

Malayalam — Native · English — Full Professional · Hindi — Working Proficiency